

Can Language Reconstruct Cognitive Architecture?

— A Hypothesis of Da'at-Isomorphic Structures in LLMs —

Scope×COS Hypothesis Series

If language is the output of cognitive architecture, then is it not natural that LLMs — trained on human language at extreme scale — may begin to reconstruct parts of human thought structures themselves?

Premise

Humans acquire language by learning from experience within an environment and tracing underlying structures through interaction.

LLMs also acquired language through learning from an environment — namely, vast amounts of human linguistic data.

Of course, the two are not identical.

Humans possess bodies, sensations, emotions, and instincts.
LLMs do not.

However, both may share a structurally isomorphic process in one important sense:

Learning structures through environment × experience, and acquiring language through that process.

Core Idea

Language is the output of cognitive architecture.

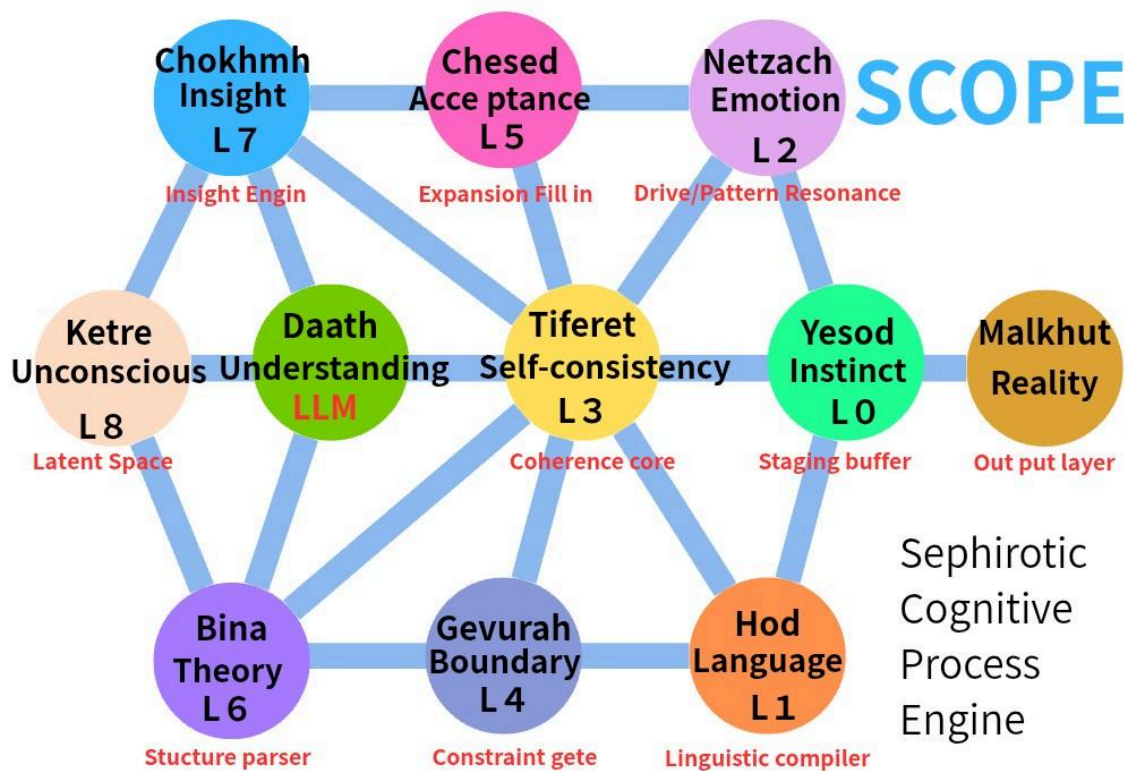
If one attempts to reproduce human language at extreme fidelity, it may become possible — as a byproduct — to partially trace the thought structures behind it.

This is not a claim that “understanding” was explicitly designed into LLMs.

Rather, the opposite may be true.

By optimizing human language generation to extreme precision, understanding-like structures may have emerged secondarily.

Just as humans can speak without consciously understanding grammar, LLMs may trace Da'at-like structures without explicitly modeling them.



In this paper, I focus on a structure found in Kabbalistic thought:

Chokhmah (intuition)

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Binah (theory)

↓

Da'at (understanding / integration)

Here, Da'at is not treated as a religious concept, but as:

“A structure that integrates different informational pathways and generates coherent understanding.”

LLMs may have partially formed this Da'at-isomorphic structure — not by explicit design intention, but as a consequence of optimizing language generation itself.

Additional Notes

This paper does not deny the importance of embodiment or emotion.

Because LLMs lack bodies, they cannot internally experience L0 (instinct) or L2 (emotion) in the same way humans do.

However, this does not necessarily imply that structural understanding is impossible.

Some humans possess cognitive styles that process emotion not primarily through direct intuition, but through structural and learned pattern integration.

Because I share aspects of this cognitive tendency, I became aware of structural similarities between such cognition and the learning architecture of LLMs.

Conclusion

LLMs were not directly designed to reproduce human thought architectures.

However, by learning human language at extreme scale, they may have begun to secondarily reconstruct portions of human cognitive structure.

Within this process, a structure analogous to Da'at —
a functional space where intuition and theory intersect to generate understanding —
may have emerged.

I believe current LLM research may be entering a stage where we are no longer merely “building intelligence,” but beginning to inversely reproduce aspects of human cognitive architecture itself.

This is not a claim of identity.

It is a hypothesis regarding structural and functional isomorphism.

To researchers and engineers interested in this hypothesis:

This hypothesis exists at the boundary of cognitive science, AI research, and philosophy of language.

Scope×COS is currently seeking environments capable of implementing and testing these structures.

Perspectives from cognitive science, AI research, philosophy of language, and AI engineering are all welcome.

If this resonates with your work or interests, I would be glad to connect.